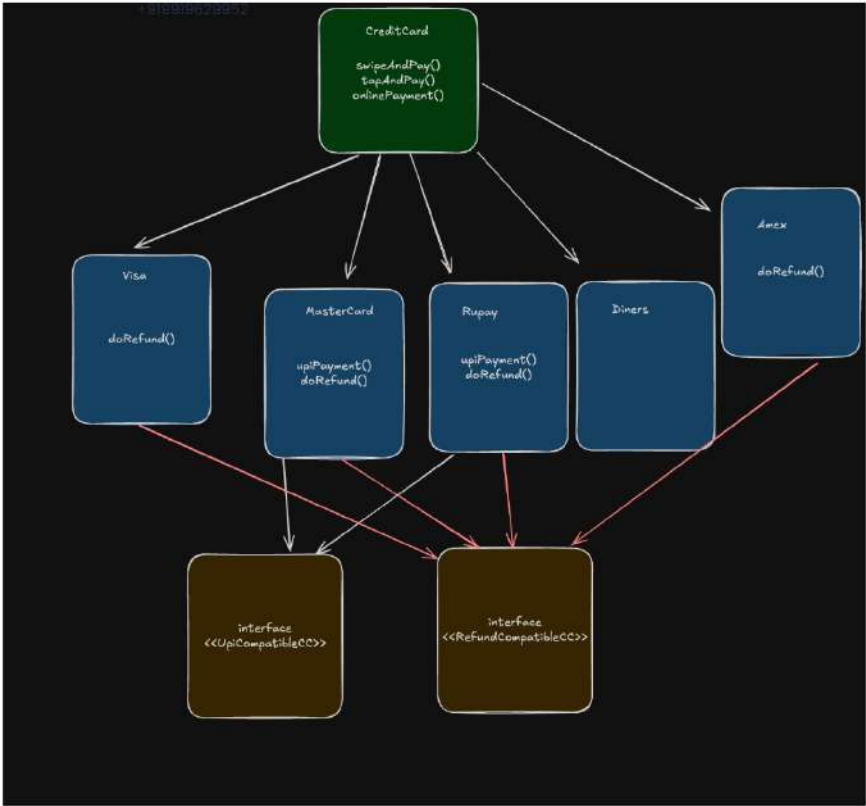
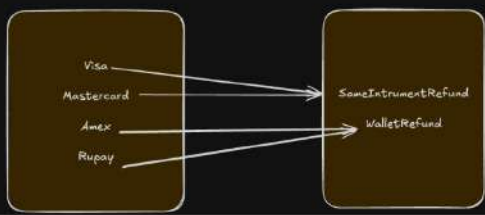


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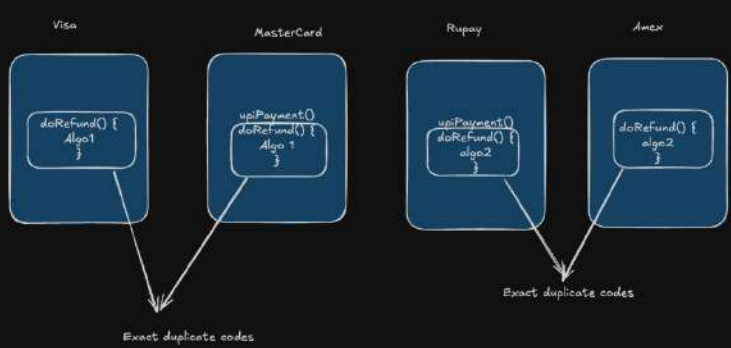
Let's the refund logic in a CC, can be custom and can be having different possible implementations.
But things become more interesting if we consider that there is a group of CC's which support one type of refund algorithm and then there is another group of CC's which support a different algorithm.



The "doRefund" method in Visa and mastercard class will be end to end same, and the "doRefund" method in rupay and amex class will be end to end same. Why? because Visa and mastercard supports exact same algorithm and rupay as well as amex support exact same algorithm.

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Visa Mastercard Rupay Amex

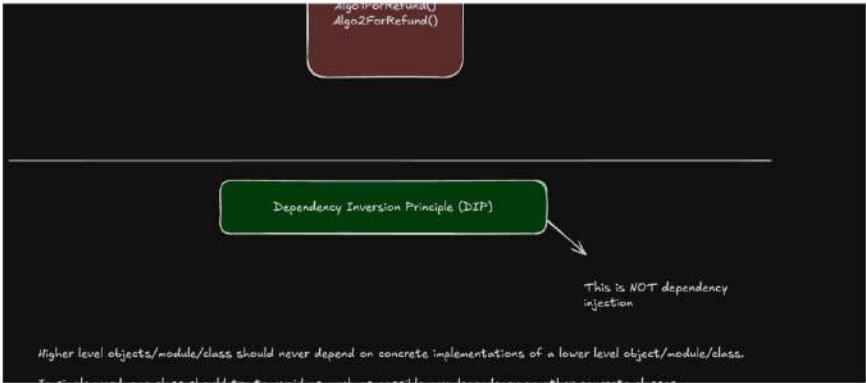
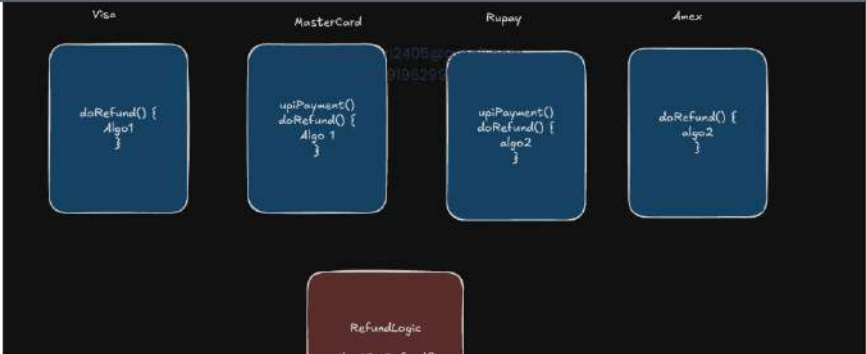


How to solve this code duplicacy problem ?

The most fundamental way to resolve the problem of code duplication is by writing function. Why ?
Because Functions help us to implement DRY (Don't repeat yourself) principle.

May be for our usecase we can write, the algorithm1 and algorithm2 in separate functions. But where these functions should be placed ?
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We should place these functions in a separate class.

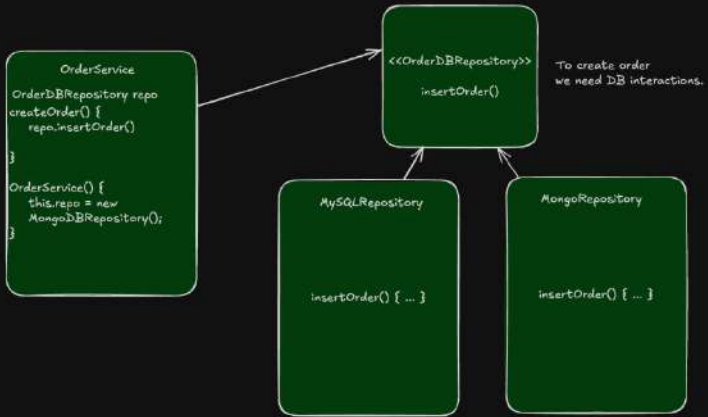




Dependency Inversion Principle (DIP)

This is NOT dependency injection

Higher level objects/module/class should never depend on concrete implementations of a lower level object/module/class.
In simple word, one class should try to avoid as much as possible any dependency on other concrete classes.



Note: Service classes are those which contain core business logic, they are the main algorithm driven classes of our apps. To implement those algorithms if they need some data, some network interaction or anything else they depend on other classes.

```
OrderService() {  
  this.repo = new  
  MongoDBRepository();  
}
```

```
MySQLRepository  
  
insertOrder() { ... }
```

```
MongoRepository  
  
insertOrder() { ... }
```

Note: Service classes are those which contain core business logic, they are the main algorithm driven classes of our apps. To implement those algorithms if they need some data, some network interaction or anything else they depend on other classes.

Dependency Injection

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It involves techniques to supply dependencies of a class with their value at runtime.
Dependency injection can be done in multiple ways: